Activity 3 Report: Cross-Group Deployment and Verification

COMP 018 Database Administration | Project tested: CampusSafe (Campus Hazard and Safety Reporting System) (PHP + PostgreSQL) | Date of testing: 7 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	CampusSafe (Campus Hazard and Safety Reporting System)
Owning group	Group 7 (member names are not listed in the project files)
Repository	grp07-campussafe (project zip grp07-campussafe-main; the repository link is not included in the files we received)
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a3_deploy.txt, screenshots

Every result below comes from this run on a freshly created database.

2. Test environment

Item	Value
Operating system	Windows 11
PostgreSQL	18.6 (x86_64 Windows), psql on PATH. The README states PostgreSQL 14+
PHP	8.2.12 CLI with pdo_pgsql, pgsql, fileinfo and session enabled
Web server	PHP built-in server (php -S localhost:8000 -t public), one of the two options the README lists
Database	grp07_campussafedb, the name in .env.example
Browser	Web browser on Windows (Figures 1 to 6)

3. Installation procedure and result

The README offers a one-click wizard (start.bat or setup.bat) and a manual guide. We followed the manual guide so that every step could be recorded.

Step	Action	Result
1	Copy .env.example to .env and set DB_NAME, DB_USER, DB_PASSWORD	Done. The template already names grp07_campussafedb and campussafe_app. We first connected as postgres, as the README says for the initial scripts
2	CREATE DATABASE grp07_campussafedb	OK
3	Run database/01_schema.sql	No errors. 9 tables, 24 indexes. A second run produced no errors either, so the

		script is repeatable
4	Run database/02_seed.sql	No errors. Roles, categories, locations, 16 reports, assignments, history and attachments loaded. A second run also gave no errors
5	Run database/seed_users.php	Creates six demo users with bcrypt hashes. Run twice: the second run updates the same users instead of duplicating them
6	Start the server and open the app	Login page shown. All six demo accounts log in with the demo password from the README
7	database/verify.sql (shipped with the project)	Row counts, status distribution and integrity checks all pass (0 violations)

The setup worked first time with no changes to the project files.

4. Database verification

Check	Result
Tables	9: role, app_user, hazard_category, campus_location, hazard_report, assignment, report_status_history, maintenance_action, attachment
Constraints	7 check, 14 foreign key, 9 primary key, 5 unique, 47 not null
Rules in the database	Status limited to the 6 lifecycle values; priority limited to Low, Medium, High, Critical; a verified report needs both verifier and time; a Resolved or Closed report needs a resolved time
Triggers	None
Seed data	5 roles, 6 users, 6 categories, 10 locations, 16 reports, 12 assignments, 60 history rows, 9 maintenance actions, 5 attachments
Owner	postgres owns all 9 tables during setup (the roles script comes in Activity 4)
Password storage	bcrypt hashes ($2y$10$, 60 characters) for all six users

Direct database tests: status "Banana" rejected by hazard_report_status_check; Resolved with no resolved time rejected by chk_resolved_has_date; priority "Urgent" rejected by hazard_report_priority_check.

5. Application checks

The pages were exercised with scripted web requests and a cookie jar per user, with database checks after the workflow steps, then looked at in the browser (Figures 1 to 6).

Test	Result
Anonymous visit to list, new report, all admin and maintenance pages	Each redirects to the login page (HTTP 302). Only the home page and login page are open
Wrong password, unknown user, empty fields, SQL text as the username	Rejected with "Invalid username or password" or "Please enter both ..." No details leaked
Login post without the CSRF token	HTTP 403
Role gating: all five account types against every page	Administrator: all admin pages and the shared pages. Maintenance: only its own two pages. Student and Faculty: shared pages only. Auditor: shared pages and Statistics only. Anything else answered HTTP 403
Student submits a report	Pass. Report #17 was created, status Submitted, and the status history gained a row

Empty description	Rejected: "Please fill out all required fields"
Category id 9999 (nonexistent)	Rejected, but the page shows the raw database message "SQLSTATE[23503] Foreign key violation … hazard_report_category_…"
Script tag and SQL text in the description	Accepted and stored as plain text (report #19). It is shown as text in the report page (Figure 4), so it is escaped on output, and the table was not dropped
Report post without the CSRF token	HTTP 403
Admin verifies report #19 as High	Pass. Redirects to the assign page
Verify with priority "Urgent"	Not accepted; the form is shown again and the report stays High
Admin assigns the report to demo_maint1	Pass
Technician logs progress, then resolves	Pass. Report #19 became Resolved with a resolved time, and the action log shows two entries (Figure 4)
Response headers and session cookie	Cache-Control: no-store. Session cookie has HttpOnly and SameSite=Lax. X-Powered-By shows the PHP version. No other security headers

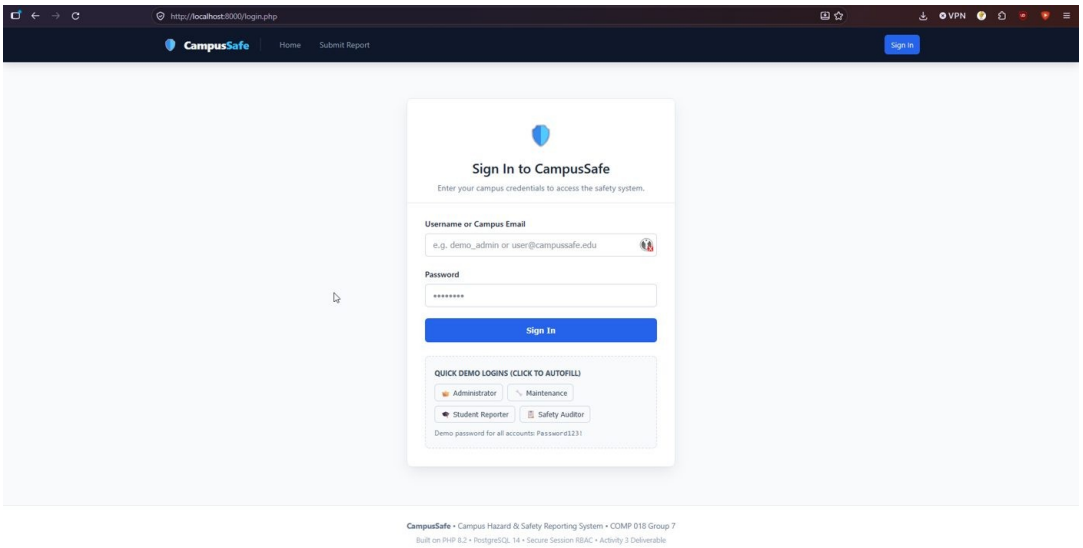


Figure 1. Login page, with the quick demo logins and the demo password shown.

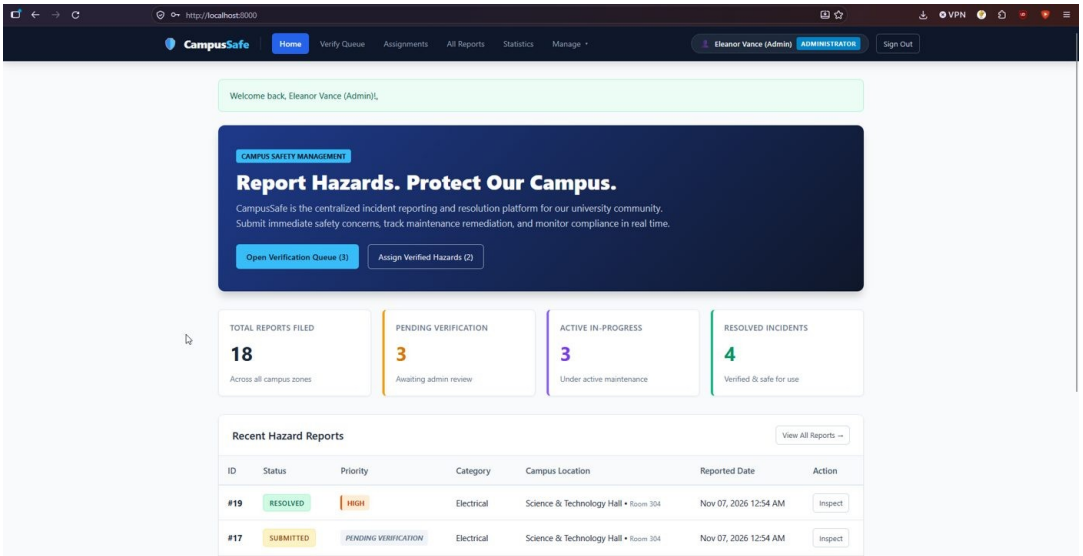


Figure 2. Administrator home page: counts and the recent hazard reports.

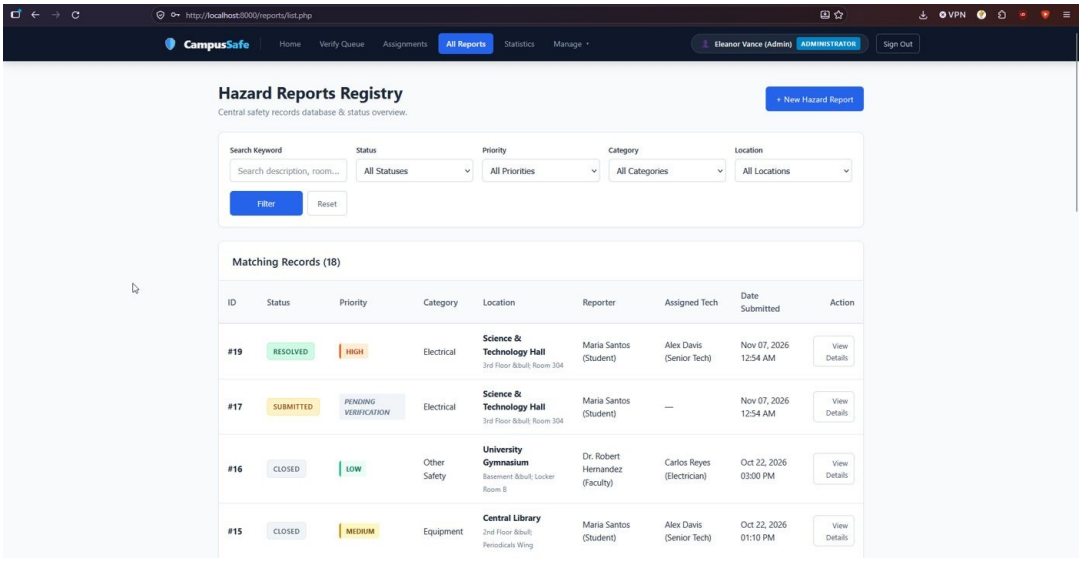


Figure 3. Hazard report list with filters. The location line shows "•" as text.

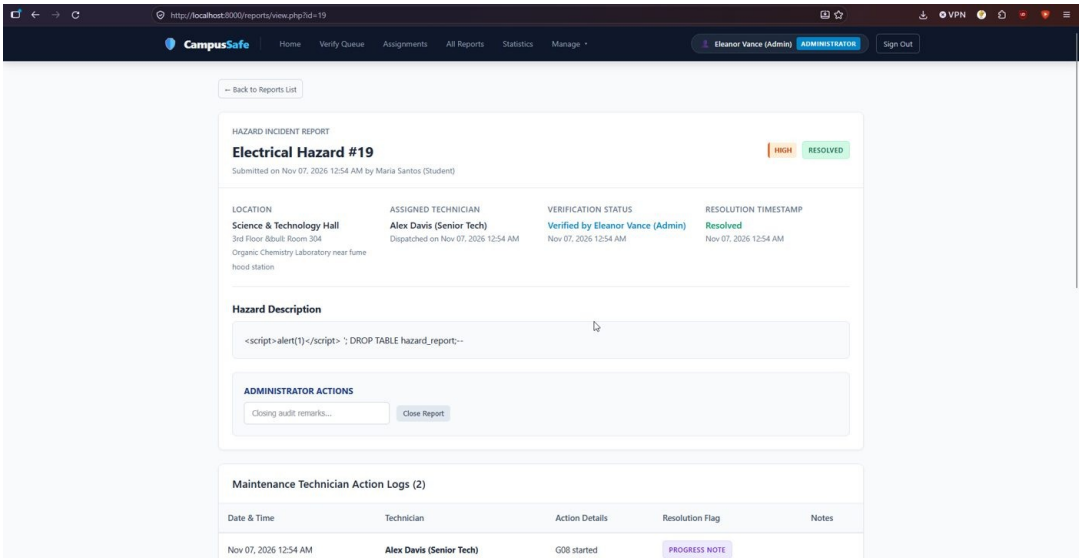


Figure 4. Report #19 after the full workflow, with the script text from the test shown as plain text.

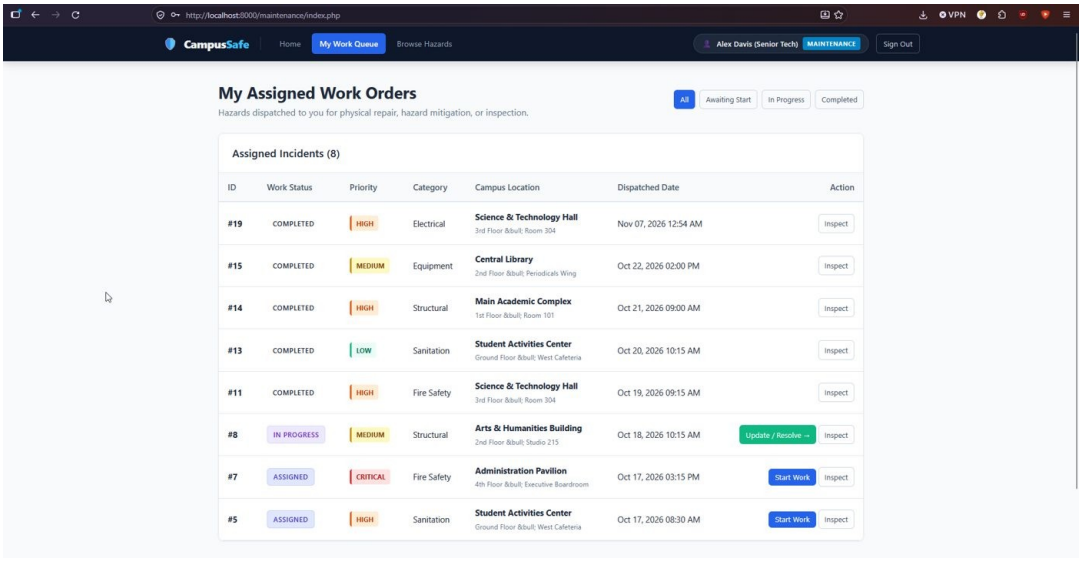


Figure 5. Technician work queue (demo_maint1).

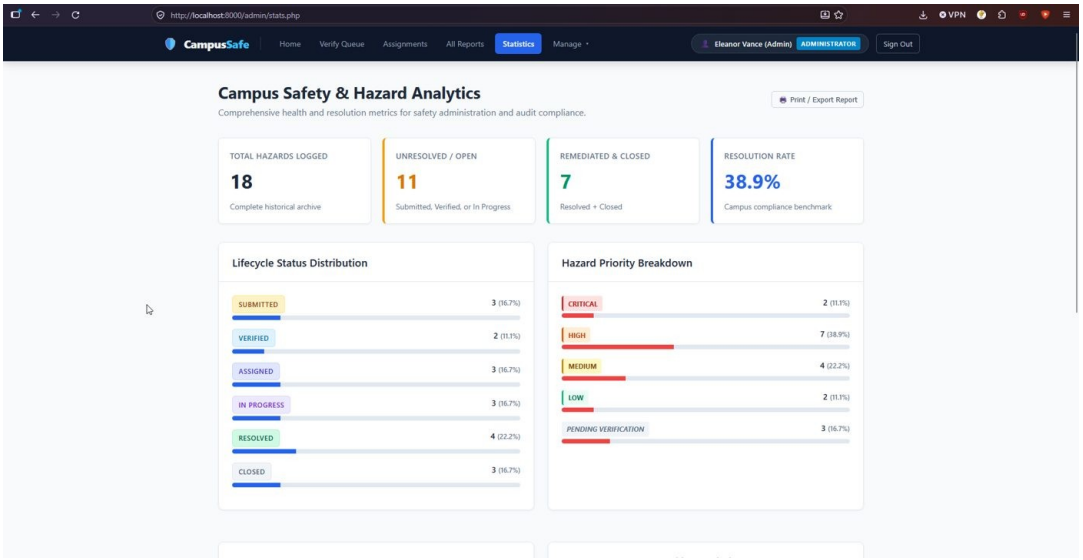


Figure 6. Statistics page: status and priority breakdown.

6. Issues found

#	Issue	Impact and suggestion
1	public/install.php is still reachable without a login after setup (it showed the installation wizard)	Anyone could reach the installer. Delete it after setup or refuse to run once the database exists
2	The login page lists the demo accounts and the demo password Password123!	Fine for a classroom demo. Remove it before any real use
3	A database foreign key error message is shown to the user when a report is filed with a bad category	Reveals table and constraint names. Check the category in PHP and show a short message
4	The location line shows a literal "•" in the list, the home page and the work queue (Figures 3 and 5)	The text is escaped twice. Escape once
5	No login lockout (see Activity 4)	Add a delay or limit after repeated failures

6	X-Powered-By header shows the PHP version	Set expose_php = Off

The lifecycle (submit, verify, assign, resolve), the role pages, the CSRF check and the database constraints all behaved as described in the README.

7. Feedback for the owners

Issues 1 and 3 are worth fixing first. The manual setup steps needed no changes.

8. Reproducibility conclusion

Reproducible. From an empty database the manual README steps gave a working system with no errors, and the workflow ran end to end.

9. Limitations

The photo and document upload feature (JPG, PNG, WEBP, PDF) was not tested in this run, only the five attachment rows that come with the seed data.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own computer, and every result in this report comes from the saved output files.